Variance and Volatility Products

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Agenda

- What are Variance Products
- Common variations
- Variance Swaps in detail
- Exotic variations
- Modeling
- References

Definitions

- **Variance products**
  - Payoff is based on realized variance
  - Variance Swap, Variance options, Capped Variance Swaps
  - VIX Futures and Options
- **Volatility products**
  - Payoff is based on realized volatility
  - Similar but not same as variance payoff
- **More exotic variations**
  - Payoff contingent on barriers
  - Vol Dispersion notes
  - Contingent variance

Realized Variance

$$\hat{\sigma}_R^2 = \frac{N}{n}\sum_{i=1}^{n} \ln\left(\frac{S_{t_i}}{S_{t_{i-1}}}\right)^2$$

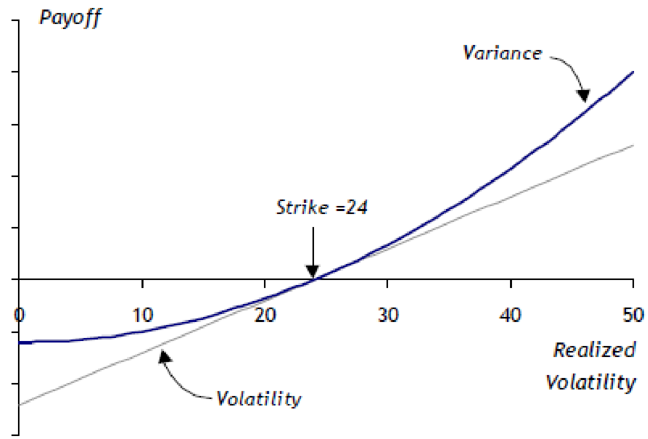
Variance Swaps

Payoff

- $V_T = \frac{100}{2K}\left(\sigma_R^2 - K^2\right)$

Convexity

- Convex in Volatility
- Higher fair strike than fair volatility
- $E\sqrt{variance} < \sqrt{E(variance)}$
- Fair strike in line with 90% put imp vol
- Quick approximation:
  – If we assume log skew:



$$K_{var} \approx \sigma_{ATMF}\sqrt{1 + 3T \times skew^2}$$

$$K_{var} \approx \sqrt{\sigma_{ATMF}^2 + \beta\sigma_{ATMF}^3 T + \frac{\beta^2}{4}\left(12\sigma_{ATMF}^2 T + 5\sigma_{ATMF}^4 T^2\right)}$$

$$\sigma(K) = \sigma_{ATMF} - \beta\ln(K/F)$$

Variance Swaps valuation -1

- **Delta Hedging Vanilla options**
  - Most material sensitivities : Delta , Gamma, Vega, Theta
  - Delta Neutral portfolio P&L

Daily P&L = Gamma P&L + Theta P&L + Vega P&L + Other

$$\text{Daily P\&L} = \frac{1}{2}\Gamma \times (\Delta S)^2 \quad + \quad \Theta \times (\Delta t) \quad + \quad \mathcal{V} \times (\Delta\sigma) \quad + \ldots$$

  - If implied vol is constant $\text{Daily P\&L} = \frac{1}{2}\Gamma \times (\Delta S)^2 \quad + \quad \Theta \times (\Delta t)$
  - From Black Scholes $\Theta \approx -\frac{1}{2}\Gamma S^2 \sigma^2$
  - Plugging we get $\text{Daily P\&L} = \frac{1}{2}\Gamma S^2 \times \left[\left(\frac{\Delta S}{S}\right)^2 - \sigma^2 \Delta t\right]$
  - Final P&L $\text{Final P\&L} = \frac{1}{2}\sum_{t=0}^{n} \gamma_t \left[r_t^2 - \sigma^2 \Delta t\right]$

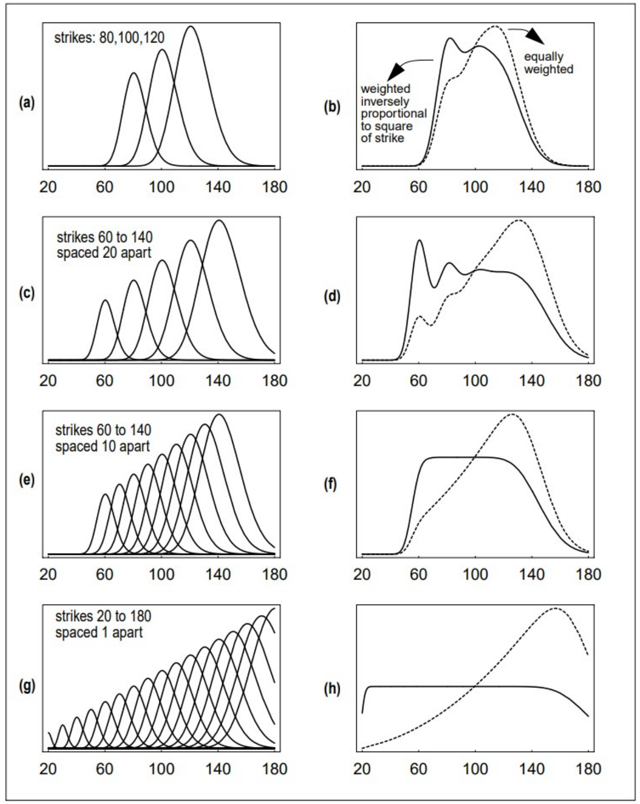
Variance Swaps valuation -Intuition

- So delta hedged option PnL, can be used as a starting point to replicate realized variance
- But we need to account for dollar gamma of our position
- How do we combine these options, so that combined portfolio has constant realized variance?
    - By constant, we mean it's not sensitive to spot move. We are looking for variance Vega of portfolio to be 0.

Here, we define the sensitivity $V = \frac{\partial C_{BS}}{\partial \sigma^2} = \frac{S\sqrt{\tau}}{2\sigma}\frac{\exp(-d_1^2/2)}{\sqrt{2\pi}}$, where $d_1 = \frac{\log(S/K) + (\sigma^2\tau)/2}{\sigma\sqrt{\tau}}$. We will sometimes refer to V as "variance vega". Note that d_1 depends only on the two combinations S/K and $\sigma\sqrt{\tau}$. V decreases extremely rapidly as S leaves the vicinity of the strike K.

- Another way to look at this is to see that we need option gamma $\propto \frac{1}{S^2}$
    - Such payoff is ln(S)
- So we can either take position in ln(S) or build a portfolio of $\frac{1}{K^2}$ weighted call options
- Turns out these both approaches are equivalent thanks to Carr Madan Formulation

FIGURE 1. The variance exposure, V, of portfolios of call options of different strikes as a function of stock price S. Each figure on the left shows the individual V_i contributions for each option of strike K_i. The corresponding figure on the right shows the sum of the contributions, weighted two different ways; the dotted line corresponds to an equally-weighted sum of options; the solid line corresponds to weights inversely proportional to K^2, and becomes totally independent of stock price S inside the strike range



Variance Swaps valuation - 2

$$p(S_T, T; S_t, t) = \left.\frac{\partial^2 \tilde{C}(S_t, K, t, T)}{\partial K^2}\right|_{K=S_T} = \left.\frac{\partial^2 \tilde{P}(S_t, K, t, T)}{\partial K^2}\right|_{K=S_T}$$

$$\begin{aligned}\mathbb{E}\left[g(S_T)\middle| S_t\right] &= \int_0^\infty dK\, p(K, T; S_t, t)\, g(K) \\ &= \int_0^F dK\, \frac{\partial^2 \tilde{P}}{\partial K^2}\, g(K) + \int_F^\infty dK\, \frac{\partial^2 \tilde{C}}{\partial K^2}\, g(K)\end{aligned}$$

$$\begin{aligned}\mathbb{E}\left[g(S_T)\middle| S_t\right] &= \left.\frac{\partial \tilde{P}}{\partial K}\, g(K)\right|_0^F - \int_0^F dK\, \frac{\partial \tilde{P}}{\partial K}\, g'(K) + \left.\frac{\partial \tilde{C}}{\partial K}\, g(K)\right|_F^\infty \\ &\quad - \int_F^\infty dK\, \frac{\partial \tilde{C}}{\partial K}\, g'(K) \\ &= g(F) - \int_0^F dK\, \frac{\partial \tilde{P}}{\partial K}\, g'(K) - \int_F^\infty dK\, \frac{\partial \tilde{C}}{\partial K}\, g'(K) \\ &= g(F) - \left.\tilde{P}(K) g'(K)\right|_0^F + \int_0^F dK\, \tilde{P}(K)\, g''(K) \\ &\quad - \left.\tilde{C}(K) g'(K)\right|_F^\infty + \int_F^\infty dK\, \tilde{C}(K)\, g''(K) \\ &= g(F) + \int_0^F dK\, \tilde{P}(K)\, g''(K) + \int_F^\infty dK\, \tilde{C}(K)\, g''(K)\end{aligned}$$

Example of Replication

Amortizing contract

$$g(S_T) = \frac{(S_T - L)^+}{S_T}$$

$$g''(K) = \left\{ -\frac{2L}{{S_T}^3}\,\theta(S_T - L) + \frac{\delta(S_T - L)}{S_T} \right\}\Bigg|_{S_T=K}$$

$$\mathbb{E}\left[\frac{(S_T - L)^+}{S_T}\right] = \int_F^\infty dK\,\tilde{C}(K)\,g''(K)$$

$$= \frac{\tilde{C}(L)}{L} - 2L\int_L^\infty \frac{dK}{K^3}\,\tilde{C}(K)$$

Log Contract

Now consider a contract whose payoff at time T is $\log(S_T/F)$. Then $g''(K) = -\left.1/{S_T}^2\right|_{S_T=K}$ and it follows from equation (11.1) that

$$\mathbb{E}\left[\log\left(\frac{S_T}{F}\right)\right] = -\int_0^F \frac{dK}{K^2}\,\tilde{P}(K) - \int_F^\infty \frac{dK}{K^2}\,\tilde{C}(K)$$

Rewriting this equation in terms of the log-strike variable $k := \log(K/F)$, we get the promising-looking expression

$$\mathbb{E}\left[\log\left(\frac{S_T}{F}\right)\right] = -\int_{-\infty}^0 dk\,p(k) - \int_0^\infty dk\,c(k)$$

For 0 Rate and no dividend

$$\begin{aligned}\log\left(\frac{S_T}{F}\right) &= \log\left(\frac{S_T}{S_0}\right)\\ &= \int_0^T d\log(S_t)\\ &= \int_0^T \frac{dS_t}{S_t} - \int_0^T \frac{{\sigma_{S_t}}^2}{2}dt\end{aligned}$$

Variance Swaps valuation

$$\mathbb{E}\left[\int_0^T \sigma_{S_t}^2 dt\right] = -2\,\mathbb{E}\left[\log\left(\frac{S_T}{F}\right)\right]$$
$$= 2\left\{\int_{-\infty}^{0} dk\, p(k) + \int_0^{\infty} dk\, c(k)\right\}$$

$$c(y) := \frac{\tilde{C}(Fe^y)}{Fe^y};\ p(y) := \frac{\tilde{P}(Fe^y)}{Fe^y}$$

Variance Swap Payoff: N(notional), A(Annualization Factor)

$$N \times A \times \left\{\frac{1}{N}\sum_{i=1}^{N}\left\{\log\left(\frac{S_i}{S_{i-1}}\right)\right\}^2 - \left\{\frac{1}{N}\log\left(\frac{S_N}{S_0}\right)\right\}^2\right\} - N \times K_{var}$$

$$VarSwap_0 \approx \frac{2}{T}\left[\sum_{i=1}^{N^{puts}}\frac{p_0(k_i^{put})}{(k_i^{put})^2}\left(k_i^{put} - k_{i-1}^{put}\right) + \sum_{i=1}^{N^{calls}}\frac{c_0(k_i^{call})}{(k_i^{call})^2}\left(k_i^{call} - k_{i-1}^{call}\right)\right]$$
$$- PV_0(T) \times (K^{VS})^2$$

Variance Swaps- variations

- Capped variance swaps
  - Cap on realized variance
  - Payoff is $\frac{100}{2K}(RV - K^2)$
  - Where RV is now $\frac{252}{N}\sum_{i=1}^{N}\left[\min(\max\left(ln\frac{S_i}{S_{i-1}}, f\right), c\right]^2$
  - Can be decomposed as Capped + Uncapped part
  - RV = $\frac{252}{N}\sum_{i=1}^{N}(r_i^2 + \left\{f^2 - r_i^2\right\}_{r_i<f} + \left\{c^2 - r_i^2\right\}_{r_i>c})$
- Corridor variance swap
  - Variance accumulates on days on which spot is within a corridor i.e lower and upper barriers
  - Payoff is again $\frac{100}{2K}(RV - K^2)$
  - But RV is $\frac{252}{N}\sum_{i=1}^{N} w_i ln\frac{S_i}{S_{i-1}}$ where $w_i = 1_{L<S_t\leq U}$ for T type monitoring
  - Similarly we have T-1 type and T and T-1 type as well
- Gamma variance swap
  - Daily variance weighted by spot performance
  - RV is $\frac{252}{N}\sum_{i=1}^{N} w_i ln\frac{S_i}{S_{i-1}}$ where $w_i = S_t$ for T type monitoring
  - Similarly, we can define T-1 type as well

Volatility Swaps and Variance Option

- **Volatility Swap**
  - $V_T = 100 * (\min\{\sigma_R, cK\} - K)$
  - c is an optional cap
  - Returns can be div adjusted

- **Variance option**
  - $V_T = \frac{100}{2\, K^{Ref}} * \max(0, p{\sigma_R}^2 - pK^2)$
  - p is1,-1 for call and put
  - Returns can be div adjusted

VIX based payoff

- **VIX**

$$\left(\frac{\mathrm{VIX}}{100}\right)^2 = \frac{2}{\tau}\sum_i \frac{\Delta K_i}{K_i^2} Q_i(K_i) - \frac{1}{\tau}\left[\frac{F}{K_0} - 1\right]^2 \approx \frac{1}{\tau}\mathbb{E}\int_0^\tau \sigma_s^2 ds$$

- **VIX Futures**
  - Future contracts based on above index traded on exchange
  - Vol Tenor 30 days in most cases
  - Prices known from exchange , so Fair value not required
  - Modeling required for risks
  - Additive bias to match market prices.
  - Modeled using stochastic vol model

- **VIX Options**
  - Call/Put on VIX
  - Priced using Black model with input of implied vol(of VIX) amongst others

More Exotic Variance Products

- KI/KO variance swap is a variance swap subject to a single KI or KO barrier monitoring the underlying asset spot.
  - For KO barrier, the realized variance is sampled from the variance start date to the KO date or maturity date, whichever is earlier.
  - For KI barrier, the realized variance is sampled from KI date to maturity date.
  - There is beautiful symmetrical relationship that KIVaR + KOVaR = Vanilla VarSwap
  - Priced under LVSV
- Flexible contingent variance
  - Payoffs that depend on an equity component and a variance component, possibly on different underlyings

$$\Pi_T = \text{VarPayoff} \times \text{EqPayoff}$$

  - Priced under LVSV

Models used

- **Closed Form with Numerical Integration**
  - Used for Variance Swaps, corridor var swap, gamma var swaps
  - Based on vanilla calls and put prices
  - Uses numerical integration
- **Stochastic volatility model with Jumps**
  - Stochastic volatility model + jumps
  - Stoch vol model is mean reverting(similar to Heston)
  - Jumps in both spot, variance are modelled.
  - Typically calibrated to variance swaps(on the fly)
- **Local Vol Stoch vol(LVSV) model**
  - Stoch vol with overlay of Local vol
  - On the fly calibration to Vanillas
  - Mean reverting stoch vol model

Stochastic Vol or LVSV?

	Stoch Vol Model	Multi Asset LVSV
Pros	• Perfect fit to variance swap curve initial term structure (in particular including variance swap basis)	• Perfect fit to vanillas with on-the-fly calibration
Cons	• Calibration to vanillas is not accurate • Calibration to vanillas is not on-the-fly. Risk mapping is required for risk management.	• Does not calibrate to variance swap curve initial term structure. In particular variance swap basis is not included in the model.

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References

- More Than You Ever Wanted To Know About Volatility, Derman, Et al